



Using the fx-991CW for number and algebra

Task A: Conversions

1) Convert the following times to hours.

a) 1 hour and 15 mins = 1.25

b) 2 hours, 20 mins and 20 secs = 2.33 $\bar{3}$

c) 4 hours, 15 mins and 16 secs = 4.25 $\dot{4}$

2) Convert the following number of hours to hours, minutes and seconds.

a) 7.25 = 7 hours and 15 minutes

b) 3.7 = 3 hours and 42 minutes

c) 7.42 = 7 hours, 25 minutes and 12 seconds

d) 3.63 = 3 hours, 37 minutes and 48 seconds

3) Fill in the gaps.

a) $121 = \boxed{11}^2$

b) $5500 = 2^{\boxed{2}} \times \boxed{5}^5 \times 11$

c) $384 = \boxed{2}^7 \times \boxed{3}^1$

d) $351 = 3^{\boxed{3}} \times \boxed{?}^{\boxed{?}} \times 13$

The value represented by the question marks should equal 1.

This can be done by having a base of 1 and any exponent or by having any base and an exponent of 0.



Task B: Solving linear equations

1) Solve the following without using a calculator and then check your answers using the calculator.

Which way of solving is faster?

$$\text{a) } \frac{1}{2}x + 7 = 25$$

$$x = 36$$

$$\text{b) } \frac{3x - 5}{2} = \frac{2x + 1}{10}$$

$$x = 2$$

$$\text{c) } 8(y + 5) = 13 - \frac{y + 5}{2}$$

$$y = -3.47058\dots$$

$$y = -\frac{59}{17}$$

2) Solve the following pairs of simultaneous equations using your calculator.

$$\begin{aligned} \text{a) } 2a + 3b &= 1 \\ -a + 4b &= 8 \end{aligned}$$

$$a = -\frac{20}{11} \text{ or } -1\frac{9}{11}$$

$$b = \frac{17}{11} \text{ or } 1\frac{6}{11}$$

$$\begin{aligned} \text{b) } 2b + a &= 8 \\ 2b + 3 &= 7a \end{aligned}$$

$$a = \frac{11}{8} \text{ or } 1\frac{3}{8}$$

$$b = \frac{53}{16} \text{ or } 3\frac{5}{16}$$



Task C: Solving polynomials

1) Solve these quadratic inequalities (to 2 d.p. if appropriate).

a) $-x^2 - 8x + 20 \geq 0$ $-10 \leq x \leq 2$

b) $x^2 + 8x - 19 < 0$ $-9.92 < x < 1.92$

c) $x^2 + 6x + 8 < 3$ $-5 < x < -1$

d) $5x^2 - x + 5 > 3x^2 - 10x - 2$ $x < -\frac{7}{2}, -1 < x$

